DisasterM3 vs BRIGHT — Optical Data Audit & EDA

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We checked two datasets, **DisasterM3** and **BRIGHT**. We looked only at normal color photos. We left out radar (SAR) pictures.

We can use both datasets for our color-photo model. But we found two big things:

1. DisasterM3 has a **leak** in its test set.
2. Our five BRIGHT events can show the model **where buildings are**. They **cannot** show it what destroyed buildings look like.

The 7 main points

1. **There is a leak in the test set.** We have 645 test pairs. 40 of them (6.2%) are almost copies of learning pairs. The reason: two xBD events, `socal_fire` and `woolsey_fire`, have the same 264 "before" photos. These 40 pairs hold 8.6% of all destroyed pixels in the test set.
2. **Our BRIGHT tiles do not leak.** Our five events have 1,285 tiles. None of them is in the photos our model learns from or is tested on.
3. **BRIGHT has no color "after" photo.** All 4,246 BRIGHT "after" pictures are SAR. So we get only "before" photos. In a "before" photo, every building still stands, so the tiles go in as **intact**. 82% of these tiles are volcano photos, but the test set has only 8 volcano pairs.
4. **The papers give the wrong resolution.** BRIGHT's own files do not match BRIGHT's paper for 6 of 14 events. The DisasterM3 paper says 0.8 m, but its copies of BRIGHT tiles are 0.30 m.
5. **Our photos look different.** Our five events are darker (pixel middle 60, pool 84) and 2.6 times sharper than the photos the model reads.
6. **"Before" and "after" photos do not line up.** The paper says they match exactly. But in 52% of the pairs we could measure, the photos are moved by more than 2 pixels, and in a few pairs by up to 56 pixels.
7. **File names hide SAR.** 73% of DisasterM3's SAR "after" files have no "sar" in the name. The tags and the pixels are right. Only the names are wrong.

Hard words used in this report

Word	Easy meaning
Optical (color) photo	A normal color photo from a satellite.
SAR picture	A radar picture. It is black and white. We do not use it.

Pre / post photo	Pre = the photo taken **before** the disaster. Post = the photo taken **after** the disaster.
Pair	One pre photo and one post photo of the same place.
Tile	One square photo, 1024 × 1024 pixels.
Label / mask	A picture that marks every pixel: building or not, broken or not.
Intact / damaged / destroyed	Not broken / a little broken / fully broken.
Learning set (train)	The photos the model learns from: 5,798 pairs.
Test set (frozen val)	The photos we use to test the model: 645 pairs. "Frozen" means we never change it.
Leak	Some test photos are also in the learning photos. The model has seen the answers before the test.
Pool (optical pool)	The 6,443 color pairs our champion model uses (learning set + test set).
Tag	A short note saved with a file, for example "this is SAR".
Resolution (GSD, metres per pixel)	How much ground one pixel shows. 0.3 m means one pixel shows 30 cm. A smaller number means more detail.
Exact copy (byte for byte)	Two files are exactly the same. Not one bit is different.
Middle value (median)	Sort all the values. The one in the middle is the median.
IoU	A score from 0 to 1. It shows how well two masks cover the same area. 1 means a perfect match.
Destroyed IoU / the gate	How well the model finds destroyed buildings. Our target ("gate") is above 0.30.

1 What we checked, and how we removed SAR

We counted every number ourselves, on the copy of the data on our computer. We did not copy numbers from the papers.

We removed SAR by **looking at the pixels**, not only by reading the tags.

DisasterM3 has two parts. **Instruct** holds the photos for learning. **Bench** holds the photos for testing (a "benchmark").

Source	What we used	Size
DisasterM3 Instruct photos	Pre photos and color post photos (we read the pixels)	20,988 files
DisasterM3 Instruct masks	Building and road masks of the pool	6,443

DisasterM3 Bench	Only its list file (JSON). The Bench photos are **not** on our computer.	30,042 entries
BRIGHT pre-event	Color tiles, and the resolution saved in each file (GeoTIFF tag)	3,395 tiles, 11 events
BRIGHT post-event	**Not used.** All of them are SAR.	4,246 tiles
BRIGHT labels	Damage maps: 0 = no building, 1 = intact, 2 = damaged, 3 = destroyed	4,246 maps

How we find SAR (the SAR rule)

- A color photo has 3 channels: red, green and blue.
- In a SAR picture, the 3 channels are the same.
- So: if the 3 channels are almost the same (the difference is less than 1 grey level), the picture is SAR.
- We check only the pixels that have data. We skip black empty pixels.
- We also compare our result with DisasterM3's tags (`image_type`, `post_image_type`). We need the pixel test because **2,066 files have no tag**.

Pairs, the pool, and BRIGHT's photos

- **Pairs.** The pairs come from the file `train_release.json`. It has **7,159 color pairs** and **2,647 SAR pairs**. They use 9,116 pre photos. fixed
- **The pool.** Our champion model uses **6,443 color pairs**. They are split with `random_state=42`: 5,798 for learning and 645 for testing. The 645 pairs are the frozen test set of our E-protocol.
- **BRIGHT has color photos only from before the disaster.** Its folder is called `pre-event_wo_ukraine_myanmar_mexico` ("wo" means "without"). So it has no photos of Ukraine (513 tiles), Myanmar (126) and Mexico (212). These 3 events have labels and SAR, but no color photo here.

2 How many photos of each kind

With the pixel test, DisasterM3 Instruct on our computer has **17,451 color photos** and **3,537 SAR pictures**. BRIGHT adds **3,395 color pre tiles**. BRIGHT adds **no** color post photo.

Part	Pre (color)	Post (color)	Post (SAR, left out)	Notes
DisasterM3 Instruct, all files on disk	9,746	7,705	3,537	20,988 files, the same total as the paper. But the paper says 17,190 color and 3,798 SAR.
DisasterM3 Instruct, files named in the JSON list	9,116	7,159	2,647	**2,066 files are not in the list.**
DisasterM3 pool (our model)	6,443	6,443	0	5,798 learning / 645 test

DisasterM3 Bench (list only)	2,679	2,199	856	Photos not on our computer. The paper says 5,024 color and 976 SAR.
BRIGHT	3,395 here (4,246 released)	0	4,246	Ukraine (513), Myanmar (126) and Mexico (212) pre tiles are missing.

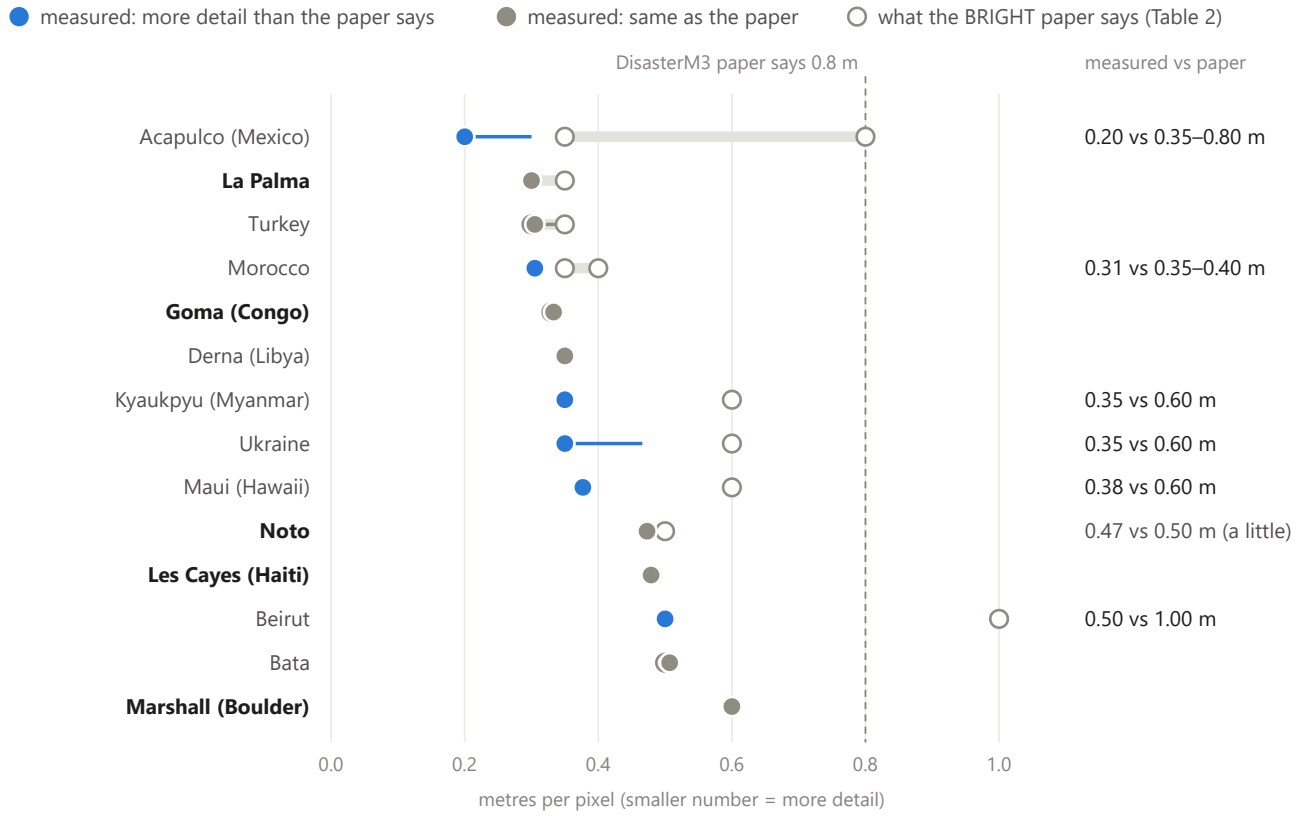
- **Tags are right. File names are not.** DisasterM3's tags and our pixel test agree on every post photo. But 2,584 of the 3,537 SAR post files (73%) have **no "sar" in the name**. Example: `la_palma_volcano_00000002_post_disaster.png` is SAR. A filter that looks at names keeps these SAR pictures as color photos. The older script `dm3_bright_audit.py` finds SAR by name, so it makes this mistake.
- **8 pre photos are more than 90% black** (1 from `rwanda_volcano`, 4 from `spain_volcano`, 3 from `ukraine_conflict`). A black pixel has red = green = blue = 0, so a test on the whole photo calls them SAR. But their few real pixels have color. This is why our rule checks only pixels with data. None of these 8 photos is in the pool.
- **2,066 files are not in any list.** Almost all of them come from BRIGHT events: `spain_volcano` 478, `la_palma_volcano` 367, Morocco under 3 spellings 617, `libya_flood` 126.
- **36 events**, the same number as the paper: 19 from xBD, 14 from BRIGHT and 3 from neither. Two BRIGHT events have new names: "Spain volcano" is La Palma, and "Rwanda volcano" is Goma. The 3 others are Kalehe, Shovi and Hurricane Ian. The paper says 10 events are new, but only these 3 are really new. 5 events are **only in Bench**: Guatemala volcano, Joplin tornado, Mexico earthquake, Hurricane Ian and Noto earthquake.

3 Resolution and photo size

BRIGHT's own files say **0.2 to 0.6 m per pixel**. DisasterM3 has 2,828 exact copies of BRIGHT tiles, and they are still **0.30 m**, not the 0.8 m that the DisasterM3 paper says.

BRIGHT tiles are 0.2–0.6 m per pixel. Six events have more detail than the paper says.

Resolution of each event, read from the files of all 4,246 BRIGHT label maps. Bold names are our five training events.



Source: dm3_bright_optical_eda.py · GeoTIFF ModelPixelScale tags (degrees changed to metres at each tile's latitude) · BRIGHT paper Table 2 · DisasterM3 paper §3

- **Beirut has the biggest gap:** 0.50 m in the files, 1.0 m in the paper.
- **Noto is a little off:** 0.47 m in the files, 0.50 m in the paper. We do not count it in the six. fixed
- **Tile size.** Every color photo in both datasets is **1024 × 1024 pixels**: 17,451 in DisasterM3 and 3,395 in BRIGHT.
- **DisasterM3's 0.8 m.** The paper (§3) says every photo was changed to 0.8 m. But its 2,828 BRIGHT copies have a middle value of 0.30 m. DisasterM3 photos have no location or scale saved in them, so we **cannot measure** the rest.

Sharpness

We measure how much small, sharp detail a photo has (the "high-frequency share"). A blurry photo scores near 0.

Photos	Sharpness (middle value)
Pool, post photos	0.073
Pool, pre photos	0.060
BRIGHT, all 11 events	0.067
BRIGHT, our 5 events	**0.189** (La Palma alone: 0.223)

Our five events are much sharper than the rest. The Turkey (0.011) and Morocco (0.009) post photos are the **most blurry** in the pool. But their pre photos are BRIGHT tiles, which are 3 to 9 times sharper.

Building size in the photo

How many pixels does one building cover? (middle value)

- In the pool: **1,030 pixels**. In all of BRIGHT: 945 pixels.
- In our five events: **471 pixels**. La Palma: 635 pixels (about 57 m^2). Goma: 422 pixels (about 47 m^2).

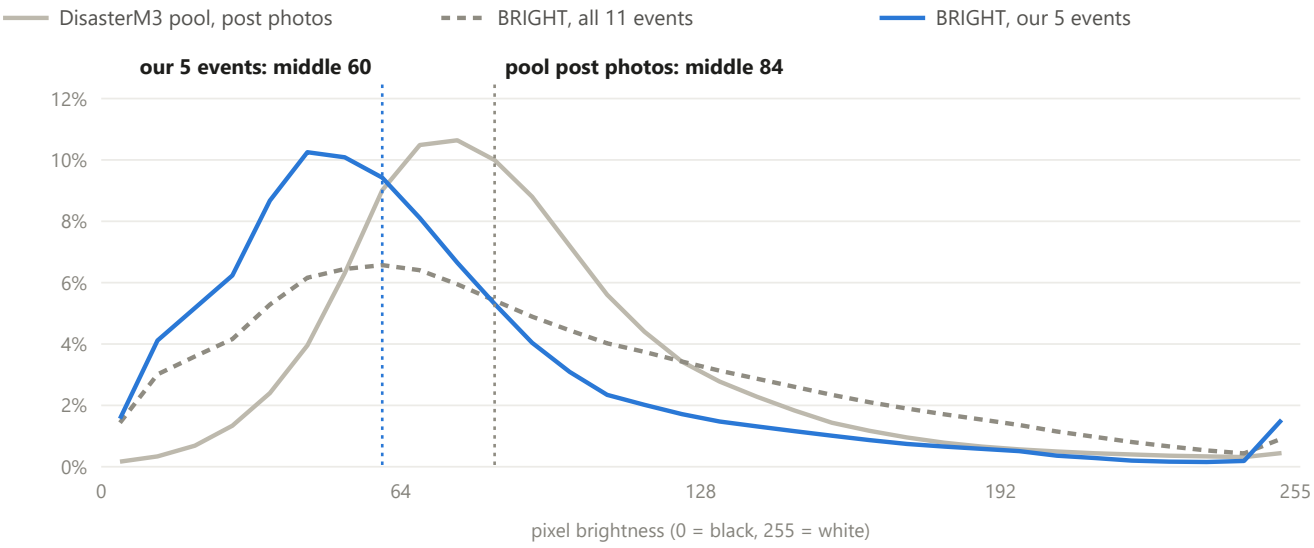
So in our five events, buildings look **small** in the photo.

4 Brightness and color

Our five BRIGHT events look the **most different** from the photos the model learns from and is tested on. They are darker, they have less color, and they are sharper. Their difference from pool post photos is 0.11 bits (0 means the same, 1 means totally different).

Our five BRIGHT events are darker than pool post photos: pixel middle 60 vs 84

Share of pixels at each brightness. Difference from pool post photos: 0.11 bits (our 5 events), 0.07 bits (all 11 events).



Source: dm3_bright_optical_eda.py · brightness of non-black pixels, over 6,443 pool post photos, 3,395 BRIGHT pre tiles and the 1,285 tiles of our five events · bins of 8 brightness levels

Most of this gap comes from La Palma. Volcano ground is dark (its middle photo brightness is 60). But DisasterM3's own events already go from 61 (Hurricane Florence) to 130 (Maui), so dark photos are not new to the model.

Two ways to measure brightness. This report uses two numbers, and they are not the same thing: fixed

- **Pixel middle: 60 vs 84.** We put all the pixels of a group together, then take the middle value.
- **Photo middle: 66 vs 89.** We find each photo's average brightness first, then take the middle of those.

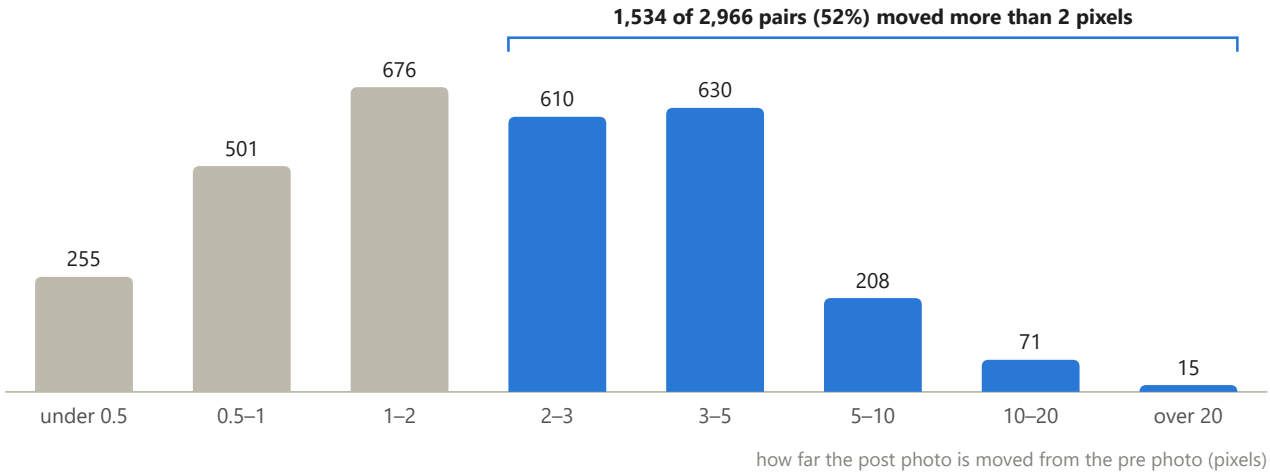
- **Pre vs post inside DisasterM3.** If we put all pre photos together and all post photos together, they look almost the same (0.002 bits). But one pair at a time, they differ: the middle difference is 0.057 bits, and 18% of pairs differ by more than 0.2 bits. The reasons are flood water, burn marks, another satellite or another season.
- **Color.** The middle color strength (saturation) is 0.30 for pool post photos and 0.21 for our five events. The photo middle brightness is 89 vs 66.
- **Black empty areas (no-data).** 10.3% of pool photos and **16.2%** of BRIGHT tiles (549 of 3,395) are more than 10% black. fixed The worst BRIGHT events are Bata and Goma (28% of their tiles each), Marshall (27%) and Derna (26%).
- **Cloud, haze and smoke.** 139 pool **post** photos are more than 20% covered by bright, colorless areas (cloud, haze or smoke). Only 15 pre photos are. So bad weather on the disaster day hits the post photos most, and the model reads the post photos.

5 Do pre and post photos line up?

In half of the pairs we could measure, the post photo is moved by **more than 2 pixels** from the pre photo (middle value 2.1 pixels). The paper says the pairs are "strictly aligned". That is not true. How far they move depends on the event.

52% of the DisasterM3 color pairs we could measure are moved by more than 2 pixels

Shift between pre and post photo, for pairs with a reliable measurement. Method: phase correlation on pieces of 256 pixels.



Source: dm3_bright_optical_eda.py · 2,966 of DisasterM3's 7,159 color pairs with a reliable shift (at least 4 pieces agree within 1.5 pixels)

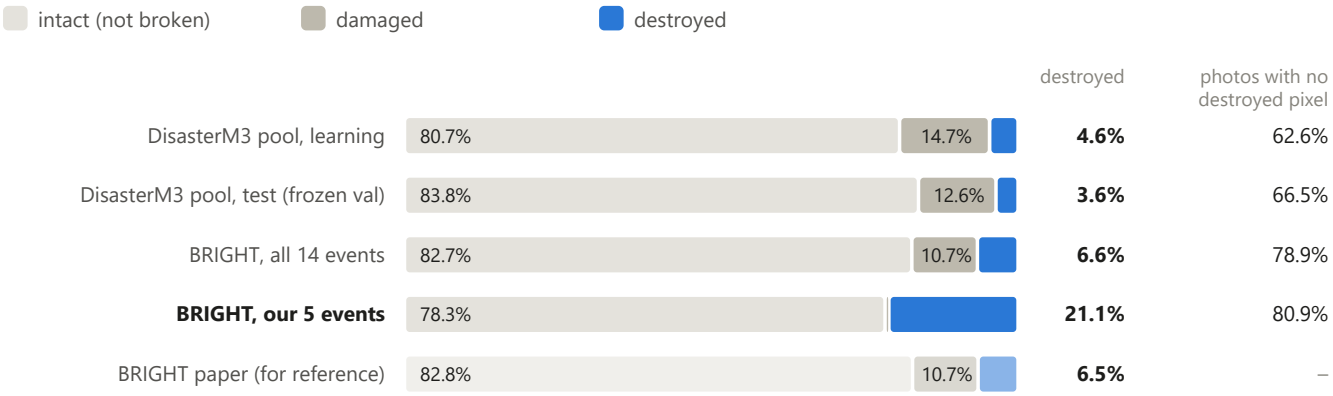
- **By event.** Portugal wildfire pairs are almost right (0.8 pixels). Hurricane Harvey is moved 7.1 pixels, Turkey scene 7 is moved 10.5 pixels and Morocco 18.7 pixels.
- **How many we could measure.** Only 2,966 of 7,159 pairs (41%). In the other pairs, real change (broken buildings), cloud or black areas stop the two photos from matching. 47 pairs have no usable piece at all.
- **Worst cases.** `turkey_earthquake1` pairs 507, 516 and 522 are moved 40–56 pixels. `hurricane_florence` pairs 107, 150, 427 and 476 are moved 26–31 pixels.
- **Who is hurt.** Our champion model reads **only the post photo**, so this problem does not hurt it. A model that reads both photos (Siamese or change detection) must **line the photos up first**, or learn with shifted copies.

6 Labels: how many of each class

Where the model is tested, destroyed buildings are **very few**: only 3.6% of building pixels in the test set are destroyed, and 66.5% of test photos have no destroyed pixel at all.

Our five BRIGHT events are 21% destroyed, but we train on them as intact

Damage class as a share of building pixels. The test set is 3.6% destroyed.



Source: dm3_bright_optical_eda.py · 4-class maps of the 6,443 pool photos (merged as the notebooks do) and the 4,246 BRIGHT label maps · BRIGHT paper Fig. 5d

BRIGHT's labels match its paper (14.1% building pixels; 82.7 / 10.7 / 6.6%). So the **labels are fine**. The problem is the **photo**: our pipeline sees only the "before" view, when every building still stands. Also, 80.9% of our tiles have no destroyed pixel, so the destroyed buildings sit in a few tiles only.

- **The merge is clean.** Our notebooks join three masks (intact, damaged, destroyed) into one map. No pixel is in two masks, and every mask holds only 0 or 255. So the order of joining does not matter.
- **Big differences between events.** Inside the pool, the destroyed share of building pixels goes from 0.07% (Turkey scene 0) to 63% (Maui).
- **Counting buildings.** The pool has 249,623 buildings (middle size 1,030 pixels). BRIGHT's maps have 211,318, but its paper says 384,596. The reason: buildings that touch are counted as one. Example: in Turkey we count 66,055, the paper says 135,033.
- **Empty tiles.** 839 BRIGHT tiles (19.8%) have no building at all. 315 of them are in our five events (24.5% of their tiles). No pool photo is empty.

- **Roads.** Pool road pixels are 92.0% intact, 6.6% flooded and 1.4% covered by debris. Our segmentation model does not use roads.

7 Events that are in both datasets

All 14 BRIGHT events are also in DisasterM3. But only **6** of them reach the pool, and 3 of those 6 (Turkey, Morocco, Derna) use **exact copies** of BRIGHT pre tiles.

BRIGHT event	Name in DisasterM3	Pool pairs (learning / test)	SAR pairs	BRIGHT pre tiles with an exact copy in DisasterM3
Turkey earthquake	`turkey_earthquake0`, 1, 3, 4, 5, 7 (6 names) fixed	557 / 64	580	583 of 1,114; 579 are pool pre photos
Morocco earthquake	`morocco_earthquake`, `morocco_earthquake1`, `morroco_earthquake1`	58 / 3	453	428 of 567; 61 are pool pre photos
Derna flood	`libya_flood`	43 / 4	51	52 of 124; 47 are pool pre photos
Beirut explosion	`beriut_explosion`	23 / 4	27	0 of 133
Bata explosion	`bata_explosion`	23 / 3	26	0 of 107
Maui wildfire	`hawaii_wildfire`	14 / 1	9	0 of 65
La Palma volcano	`la_palma_volcano`, `spain_volcano`	0 / 0	1,652	921 of 933
Goma volcano	`rwanda_volcano`	0 / 0	93	93 of 123
Marshall wildfire (Boulder)	`marshall_wildfire`	0 / 0	17	17 of 77
Les Cayes earthquake	`haiti_earthquake`	0 / 0	8	8 of 73
Noto earthquake	`noto_earthquake` (Bench only)	0 / 0	0	0 of 79
Ukraine conflict	`ukraine_conflict`	0 / 0	413	cannot check: no pre tiles on our computer
Acapulco hurricane	`mexico_hurricane`	0 / 0	112	cannot check: no pre tiles on our computer
Kyaukpyu cyclone	`myanmar_hurricane`	0 / 0	96	cannot check: no pre tiles on our computer

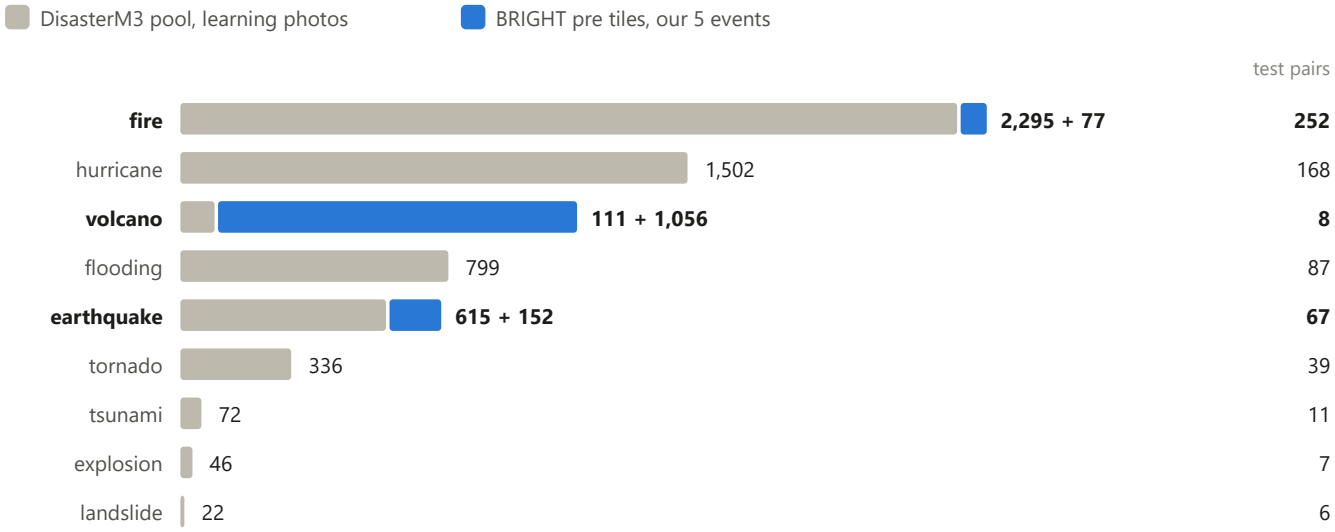
Bold: our five training events.

- **Our five events cannot leak.** None of their tiles is in the pool. Their copies in DisasterM3 are only SAR pairs and Bench masks. Two checks count these copies in two ways: fixed
  - **1,039 tiles** have an exact pixel copy in DisasterM3 (the last column of the table: 921 + 93 + 17 + 8 + 0).
  - **1,165 tiles** are found by the leak check `bright_dm3_dedup.py`, which also matches tiles by file name and building mask.

  Both counts give the same answer: **none** of these tiles is in the pool.
- **Turkey, Morocco and Libya would leak.** 66 test pairs (59 Turkey, 4 Libya, 3 Morocco) use a BRIGHT tile as their pre photo. If these tiles go into learning, the model sees test photos. So they **must stay out of training**.

BRIGHT adds 1,056 volcano tiles to learning, but the test set has only 8 volcano pairs

Learning photos by disaster type, after adding our five BRIGHT events. Right column: test pairs of each type.



Source: dm3_bright_optical_eda.py · DisasterM3 pool learning set (5,798 pairs) and test set (645) · BRIGHT pre tiles of La Palma, Goma, Les Cayes, Marshall and Noto (1,285)

Our five events add data almost only where the test **looks least**: 1,056 of their 1,285 tiles are volcano photos, and volcano has only 8 of the 645 test pairs.

The same photos twice inside DisasterM3. `la_palma_volcano` and `spain_volcano` hold the same La Palma pairs two times: 1,456 identical files, all in SAR pairs. `socal_fire` and `woolsey_fire` share 264 identical pre photos inside the pool. The next section shows what this does to the test set.

8 Checking the learning / test split

The 645-pair test set looks like the learning set: similar events and similar labels. But 40 test pairs (6.2%) show **places that are also in the learning set**, and they hold 8.6% of the test set's destroyed pixels.

Measure	Learning	Test
Pairs	5,798	645
From xBD events	87.2%	86.8%
Building pixels, out of all pixels	7.21%	7.45%
Damaged, out of building pixels	14.67%	12.63%
Destroyed, out of building pixels	4.63%	3.59%
Photos with no destroyed pixel	62.6%	66.5%

- **How the leak happens.** `socal_fire` and `woolsey_fire` are two names for the same 264 pre photos. The random split put 40 of them in the test set and their **twin** in the learning set. The twins are very alike:
  - the pre photo is exactly the same;
  - the post photos are 88% alike (correlation r = 0.88, middle value);
  - the building masks overlap at IoU 0.72;
  - 97% of the shared building pixels have the same damage class.
- **Why a name check misses it.** No test file has the same name as a learning file. Only the **pixels** match.
- **What it does to the score.** The test set has 1,808,111 destroyed pixels, and 155,812 of them are in these 40 pairs. So the 645-pair score is **a little too good**: the model gets points for remembering the twins. Report the Destroyed IoU on the **605 clean pairs** next to the 645 score. The list of the 40 pairs is in `val_train_leaks.csv`.

Every test event also appears in the learning set. This is normal for a random split. Tiles that sit next to each other on the ground could also be split between learning and test. But DisasterM3 photos have no location, so we cannot check this.

9 Problems in the data

We found 11 problems. **Only the first one** makes the test score wrong. The others change counts, filters or how we prepare photos.

#	Problem	How many	Effect on our work
1	Same place in learning and test (`socal_fire` = `woolsey_fire`)	40 test pairs, 8.6% of test destroyed pixels	The 645-pair Destroyed IoU looks **too good**
2	Pre and post photos do not line up	52% of 2,966 measured pairs over 2 px; 10% over 5 px; worst 55.6 px	None for the post-only champion. A pre+post model must handle it.
3	More than 10% of the tile is black	10.3% of pool photos, 16.2% of BRIGHT tiles fixed	Wasted crops. Brightness fixing per photo goes wrong.

4	Blurry, stretched post photos	Turkey posts have 3 times less fine detail than their pre tiles; Morocco 9 times less	Blurry post photos next to sharp 0.30 m pre tiles
5	Cloud, haze or smoke over 20% of the tile	139 pool post photos, 15 pool pre photos	It hides the damage in the photo the model reads
6	SAR files without "sar" in the name	2,584 of 3,537	Filters by name let SAR into color data
7	La Palma saved under two names (`la_palma_volcano` = `spain_volcano`)	1,456 identical files	La Palma is counted twice in DisasterM3 numbers
8	Wrong or mixed spelling of event names	`beriut` , `morroco` ; Morocco under 3 names, Turkey under 6	Grouping events by file name is wrong without a name map
9	Files not in any JSON list	2,066	Loaders that read the list never see them
10	BRIGHT tiles with no building	839 of 4,246; 315 of our 1,285	Learning tiles with only background
11	The papers give the wrong resolution	BRIGHT Table 2: 6 of 14 events (and Noto a little). DisasterM3: says 0.8 m, its BRIGHT copies are 0.30 m.	Plans based on the papers' numbers will be wrong

Also: 8 pre photos are more than 90% black, but none of them is in the pool.

10 What to do

Fix the leak first. Adding our BRIGHT tiles is safe, but **do not expect** it to raise the Destroyed score.

1. **Report the score two times.** Keep the score on all 645 test pairs, so we can compare with the old runs R1–R6. Also report the Destroyed IoU on the **605 clean pairs** (the pairs not in `val_train_leaks.csv`). If we ever make a new split, group the pairs by the fingerprint (hash) of the pre photo, so twins always stay on the same side.
2. **Keep BRIGHT Turkey, Morocco and Libya out of training.** 66 test pairs use these tiles as pre photos. Our five events are fine: none of their tiles is in the pool.
3. **Expect little change from our five events.** They add "before" photos with buildings labeled intact. 82% of them are volcano, and the test set has only 8 volcano pairs. Their labels mark 21% of building pixels as destroyed, but there is no color "after" photo of these events on our computer. The only one anywhere is Noto: 68 photos in DisasterM3 Bench.
4. **Make the photos look alike before mixing.** Our five events are darker (photo middle 66 vs 89), sharper, and their buildings are smaller (471 vs 1,030 pixels). fixed
   - Fix the brightness of each photo by itself (per-image normalization).
   - Add blur and color changes during learning (augmentation).

- Think about making these photos **1.5 times bigger**. The buildings are about 2.2 times smaller in area, and 1.5 × 1.5 = 2.25, so this makes the sizes match.

5. **Find SAR by tag or pixels, never by file name.** Stop using the name-based SAR check in `dm3_bright_audit.py` and `dm3_vs_bright_compare.py`.
6. **Skip crops that are mostly black.** 10–16% of tiles are more than 10% black. When we cut a random crop, we throw it away if more than half of it is black.
7. **Line up pairs before any pre+post model.** The champion reads only post photos. But a Siamese model would see a middle shift of 2 pixels, and more than 5 pixels in 1 of every 10 pairs.
8. **Take resolution from the file tags, not from the papers.** BRIGHT's Table 2 is wrong for 6 events. DisasterM3's BRIGHT copies are 0.30 m, not 0.8 m.

11 How to make every number again

Two commands make every number in this report again. They take about 35 minutes on a laptop with 8 GB of memory.

```
cd Essential_task_delegation/BRIGHT_DisasterM3_Integration
python dm3_bright_optical_eda.py   # the main check; results go to data/dm3_bright_optical_eda/
python bright_dm3_dedup.py         # checks every BRIGHT tile against the pool
```

Output file	One row is
`summary.json`	every number in this report
`dm3_images.csv`	one DisasterM3 Instruct photo: pre or post, event, tag, color or SAR by pixels, brightness, fingerprint (md5), split
`bright_pre_tiles.csv`	one BRIGHT pre tile: resolution, brightness, fingerprint
`bright_targets.csv`	one BRIGHT label map: pixels of each class, buildings, resolution
`dm3_pool_labels.csv`	one pool photo: pixels of each class, buildings, road pixels
`dm3_optical_pair_registration.csv`	one color pair: shift, how many pieces agree, can we trust it
`dm3_pool_pairs_radiometry.csv`	one pool pair: pre and post brightness, color, difference
`dm3_events.csv`	one DisasterM3 event name: disaster type, where it came from, counts
`val_train_leaks.csv`	one leaked test pair: its twin in the learning set, and how alike they are

What the measures mean

- **High-frequency share (sharpness):** how much fine detail is in the middle 512 × 512 pixels of a photo (the power between half and full Nyquist frequency). Blurry or stretched photos score near 0.
- **JS divergence (difference in bits):** how different two brightness patterns are. 0 means the same, 1 means totally different.
- **Reliable shift:** we cut both photos into pieces of 256 pixels and measure how far each piece moved (phase correlation). We trust the result only if at least 4 pieces give a clear signal (peak 0.1 or more) and agree within 1.5 pixels.
- **Building count (components):** a group of building pixels that touch each other. Buildings that touch are counted as one, so busy areas get **too few**.

Part 2 — Mistakes in the older PDF

The older PDF (*EO Data Audit & Exploratory Data Analysis: DisasterM3 vs. BRIGHT*, 4 pages) copied the score tables from the papers **correctly**. But **most of its data numbers are wrong**, or they do not match the files on our computer.

The older PDF said	Verdict	The truth (measured, unless we say "paper")
Cover: "4,246 BRIGHT pre-event EO tiles"	Wrong for the release	3,395 color pre tiles. The other 851 (Ukraine, Myanmar, Mexico) are Google Earth photos. You must download them yourself with `aoi_reference_kmz.zip` and BRIGHT's guide.
Cover: BRIGHT "0.3–1.0 m GSD"	Wrong	0.2–0.6 m in the file tags
Cover: "22,214 DisasterM3 EO images (0.8 m)"	Number from the paper; 0.8 m wrong	On our computer: 17,451 color photos (the paper says 17,190). Its 2,828 BRIGHT copies are 0.30 m.
Cover: "11,107 pure EO bi-temporal pairs"	Wrong	The older PDF's own math gives 17,440 / 2 = 8,720. The JSON lists 7,159 color pairs (Instruct) + 2,199 (Bench) = 9,358.
Cover: "384,596 audited building footprints"	Paper number, not checked	The paper's own rows add up to 358,340. The label maps have 211,318 building groups (touching buildings count as one).
§1: BRIGHT "5-event T1+T2 EO subset"	Not in the release	Used only in one experiment in the BRIGHT paper (§4.4). All 4,246 released post tiles are SAR.
§1: DisasterM3 "all scenes resampled to 0.80 m"	Wrong	2,828 DisasterM3 photos are exact BRIGHT copies at 0.30 m.
§1: DisasterM3 "georeferenced to 0.8 m grid"	Wrong	DisasterM3 photos have no location saved in them.
§1, §3: "xBD 15 + BRIGHT 11 + 10 new events (incl. Nepal flooding)"	Wrong	19 xBD + all 14 BRIGHT (two renamed) + 3 new: Kalehe, Shovi, Hurricane Ian. Nepal flooding is an xBD event.
§1: DisasterM3 "instance masks"	Not exact	One black-and-white (0/255) mask per class and photo. Single buildings can only be found by grouping touching pixels.
§2: resolution (GSD) column	Wrong for 6 of 14 events, a little off for Noto fixed	Beirut 0.50 (not 1.00), Ukraine 0.35–0.47 (not 0.60), Myanmar 0.35 (not 0.60), Maui 0.38 (not 0.60), Morocco 0.31 (not 0.35–0.40), Acapulco 0.20–0.30 (not 0.35–0.80); a little off: Noto 0.47 (not 0.50)
§2: color tiles for Ukraine 513, Myanmar 126, Acapulco 212	Not released	0 in `pre-event.zip`. DisasterM3 has 410, 96 and 112 of them, with BRIGHT's labels.
§2: "Reg. RMSE" column	This is SAR content	It is about lining up color photos with SAR (BRIGHT Table C1). For color pre/post pairs in DisasterM3: 52% of measured pairs are moved more than 2 pixels.

§2: "5 events (488 tiles)" with post-event color photos	**Wrong sum, and not released**	133 + 107 + 65 + 124 + 79 = 508 tiles. No color post photo is released.
§3: BRIGHT tiles "downsampled to 0.80 m", a 2.28–2.67 times loss	**The idea is wrong**	DisasterM3 keeps them as exact copies at 0.30 m (La Palma, Goma, Les Cayes, Marshall, Turkey, Morocco, Derna).
§4: BRIGHT 85.9 / 14.1%, 6.1 : 1 (paper about 7 : 1)	**Right**	14.14% building pixels, 6.07 : 1
§4: BRIGHT 82.8 / 10.7 / 6.5%	**Right**	82.7 / 10.7 / 6.6%
§4: DisasterM3 92.85 / 1.81 / 5.34%, "damaged collapses 5.9 times"	**The masks say otherwise**	Pool: 81.0 / 14.5 / 4.5% of building pixels (5.86 / 1.05 / 0.33% of all pixels, as in our Integration Plan). DisasterM3 actually has **more** damaged than BRIGHT.
§4: roads 79.44 / 12.94 / 7.62%	Paper number	Pool road pixels: 92.0 / 6.6 / 1.4%
§4: 27,074 vs 26,988 "due to 86 multi-hazard tiles"	**No proof**	The paper gives no reason. 86 is only the difference between the two numbers.
§4: photo counts by disaster type (6,750 fire …)	**Right (paper Fig. 2)**	But they mix color + SAR, and Instruct + Bench. Our color pool only: fire 2,547, hurricane 1,670, flooding 886, earthquake 682, tornado 375, volcano 119, tsunami 83, explosion 53, landslide 28, conflict 0.
§5A: BRIGHT Table 7 gains (F1 68.71 → 73.59 …)	**Numbers right, but SAR**	Table 7 uses SAR post pictures, so these are color + SAR results, not color only.
§6: BRIGHT Table 8, DisasterM3 Tables 2, 3, 9	**Right**	Every number matches the papers. One label is wrong: the BRIGHT rows put "Intact IoU" under "Counting".
§7: +3.80 mIoU, +8.32 Destroyed IoU, +5.7 mIoU, +7.9 Building IoU, 5.4 times fewer parameters	**Right**	Checked against BRIGHT Table 8 and DisasterM3 Table 3.
Pages 2–4	**Text is cut**	Sections 3, 5A and 7 stop in the middle of a sentence.

What the older PDF did not have

- The test-set leak: 40 pairs whose `socal_fire` / `woolsey_fire` twins are in the learning set, with 8.6% of the test destroyed pixels.
- All 14 BRIGHT events are inside DisasterM3, and BRIGHT's Turkey, Morocco and Libya tiles are the pre photos of 66 test pairs.
- 73% of DisasterM3's SAR post files have no "sar" in the name.
- Pre and post photos do not line up, and the BRIGHT photos look different.

How to fix the older PDF

1. Replace sections 1–5 with the numbers we measured. Put "the paper says" and "we measured" in two separate columns.

2. Keep sections 6–7, and fix the wrong column label.
3. Remove the SAR content (Table 7, Table C1), or label it "color + SAR".
4. Next to every number from a paper, write its table or figure number.
5. Export the PDF again, so no section is cut.

What we fixed in this version (Oct 22, 2026)

We checked every number against `summary.json` and `bright_dm3_dedup_report.json`. These places changed from the Oct 19 version. They are marked fixed in the text.

Where	Oct 19 version	This version
§1, Pairs	7,160 color and 2,648 SAR pairs, 9,117 pre photos	7,159 color and 2,647 SAR pairs, 9,116 pre photos. The data and the rest of the report already used these numbers.
§4 and §9, black areas	15.9% of BRIGHT tiles	16.2% (549 of 3,395 tiles)
§3 and Part 2, Morocco resolution	0.30 m in the chart, 0.31 m in Part 2	0.31 m in both places (the files say 0.305 m)
§3 and Part 2, Noto resolution	Part 1 called it a match; Part 2 called it wrong ("7 of 14")	The same rule in both parts: 6 events are clearly wrong, and Noto is a little off (0.47 vs 0.50 m)
§4 and §10, brightness	"pixel median 60 vs 84" and "median luminance 66 vs 89" with no explanation	Two clear names: **pixel middle** (60 vs 84) and **photo middle** (66 vs 89)
§7, our five events in DisasterM3	Said 1,165 tiles, while the table adds up to 1,039, with no reason given	Both counts explained: 1,039 exact pixel copies; 1,165 found by the leak check, which also matches names and masks
§7, Turkey names	`turkey_earthquake0` – 7	6 names: 0, 1, 3, 4, 5, 7. There is no 2, and 6 is only in Bench.

No conclusion changed. The leak, the five-event result and the eight recommendations are the same.